

MAT 528 HW10

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Problem 1 (Munkres 51.2). Given spaces X and Y , let $[X, Y]$ denote the set of homotopy classes of maps of X into Y .

- (a) Let $I = [0, 1]$. Show that for any X , the set $[X, I]$ has a single element.
- (b) Show that if Y is path connected, the set $[I, Y]$ has a single element.

Solution. (a) Each continuous map $f: X \rightarrow I$ is homotopic to the zero map via the homotopy

$$\begin{aligned} F: X \times I &\rightarrow I \\ F(x, t) &= (1 - t)f(x). \end{aligned}$$

We can see that $F(x, 0) = (1 - 0)f(x) = f(x)$ and $F(x, 1) = (1 - 1)f(x) = 0$ for all $x \in X$, so to verify that F is a homotopy it remains to check continuity of F ; this follows from the continuity of the maps $(x, t) \mapsto f(x)$ and $(x, t) \mapsto 1 - t$ and Munkres's Theorem 21.5.

(b) Let $f, g: I \rightarrow Y$ be two paths in Y . Let $f(0) = x_0$, $f(1) = x_1$, $g(0) = y_0$, and $g(1) = y_1$. Since Y is path connected, there exists a path $h: I \rightarrow Y$ from x_1 to y_0 . Define $F: I \times I \rightarrow Y$ by

$$F(x, t) = \begin{cases} f(x + 2t) & \text{if } x + 2t \leq 1 \\ h(x + 2t - 1) & \text{if } 1 \leq x + 2t \leq 2 \\ g(x + 2t - 2) & \text{if } x + 2t \geq 2. \end{cases}$$

This map is continuous by the pasting lemma, as $f(x + 2t)$, $h(x + 2t - 1)$, and $g(x + 2t - 2)$ are continuous maps on $\{(x, t) \in I \times I : x + 2t \leq 1\}$, $\{(x, t) \in I \times I : 1 \leq x + 2t \leq 2\}$, and $\{(x, t) \in I \times I : x + 2t \geq 2\}$, respectively, and the maps agree when these sets intersect because $f(1) = h(0) = x_1$ and $h(1) = g(0) = y_0$. Furthermore $F(x, 0) = f(x)$ and $F(x, 1) = g(x)$ for all $x \in I$, so F is a homotopy from f to g . Thus any $f, g: I \rightarrow Y$ are homotopic, so $[I, Y]$ has only one element. $\square$

Problem 2 (Munkres 51.3). A space X is **contractible** if the identity map $i_X: X \rightarrow X$ is nullhomotopic.

- (a) Show that I and $\mathbb{R}$ are contractible.
- (b) Show that a contractible space is path connected.

Solution. (a) Let $X = I$ or $\mathbb{R}$ and consider the map $F: X \times I \rightarrow X$ given by $F(x, t) = (1 - t)x$. Then $F(x, 0) = (1 - 0)x = x$ and $F(x, 1) = (1 - 1)x = 0$ for all $x \in X$, and F is continuous because it is the product of continuous maps $(x, t) \mapsto x$ and $(x, t) \mapsto t$, so F is a homotopy between the constant map and the zero map (this is the same logic as part (a) of the previous problem).

(b) Suppose i_X is homotopic to the constant map f that is equal to $c \in X$ on all of X . Then, for any $x_1, x_2 \in X$, define $f_1: I \rightarrow X$ and $f_2: I \rightarrow X$ by

$$f_1(t) = F(x_1, t), \quad f_2(t) = F(x_2, 1 - t)$$

so f_1 is a path from x_1 to c and f_2 is a path from c to x_2 ; then $f_1 * f_2$ is a path from x_1 to x_2 . $\square$

Problem 3 (Munkres 52.3). Let x_0 and x_1 be points of the path-connected space X . Show that $\pi_1(X, x_0)$ is abelian if and only if for every pair α and β of paths from x_0 to x_1 , we have $\hat{\alpha} = \hat{\beta}$.

Solution. By Munkres's Theorem 52.1, $\hat{\alpha}$ and $\hat{\beta}$ are isomorphisms $\pi_1(X, x_0) \rightarrow \pi_1(X, x_1)$, so $\hat{\beta}^{-1} \circ \hat{\alpha}: \pi_1(X, x_0) \rightarrow \pi_1(X, x_0)$, which is equal to the map

$$[f] \mapsto [\beta * \bar{\alpha}][f][\alpha * \bar{\beta}]$$

(note that the right-hand side is a product of three elements of $\pi_1(X, x_0)$). If $\pi_1(X, x_0)$ is abelian, then this composed map sends $[f] \in \pi_1(X, x_0)$ to $[\beta * \bar{\alpha}][f][\alpha * \bar{\beta}] = [\beta * \bar{\alpha}][\alpha * \bar{\beta}][f] = [f]$, i.e. the inverse of $\hat{\beta}$ is equal to the inverse of $\hat{\alpha}$, which implies $\hat{\alpha} = \hat{\beta}$.

On the other hand suppose $\hat{\alpha} = \hat{\beta}$ for any two paths α, β from x_0 to x_1 . Let $[f], [g] \in \pi_1(X, x_0)$ and β an arbitrary path from x_0 to x_1 . Define $\alpha = g * \beta$, which is also a path from x_0 to x_1 , so that $[\alpha * \bar{\beta}] = [g]$ (hence also $[\beta * \bar{\alpha}] = [\bar{g}]$). Then, by assumption, $\hat{\alpha} = \hat{\beta}$, so the two maps share an inverse, meaning

$$[f] = \hat{\beta}^{-1} \hat{\alpha}([f]) = [\beta * \bar{\alpha}][f][\alpha * \bar{\beta}] = [\bar{g}][f][g] \iff [g][f] = [f][g],$$

hence $\pi_1(X, x_0)$ is abelian. $\square$

Problem 4 (Munkres 52.4). Let $A \subseteq X$; suppose $r: X \rightarrow A$ is a continuous map such that $r(a) = a$ for each $a \in A$. If $a_0 \in A$, show that

$$r_*: \pi_1(X, a_0) \rightarrow \pi_1(A, a_0)$$

is surjective.

Solution. For each path f in A based at a_0 , the path homotopy class $[f]$ in $\pi_1(X, a_0)$ gets sent by r_* to $r_*([f]) = [r \circ f] = [f] \in \pi_1(A, a_0)$, where $r \circ f = f$ because r is the identity on A , so each $[f] \in \pi_1(A, a_0)$ is mapped to by the corresponding path homotopy class $[f]$ in $\pi_1(X, a_0)$. $\square$

Problem 5 (Munkres 52.5). Let A be a subspace of $\mathbb{R}^n$ and let $h: (A, a_0) \rightarrow (Y, y_0)$ be a basepoint-preserving continuous map. Show that if h is extendable to a continuous map of $\mathbb{R}^n$ into Y , then h_* is the trivial homomorphism.

Solution. Let $\tilde{h}: (\mathbb{R}^n, a_0) \rightarrow (Y, y_0)$ be a continuous extension of h and let $i: (A, a_0) \rightarrow (\mathbb{R}^n, a_0)$ be the inclusion map. Then $\tilde{h} \circ i = h$, so $h_* = \tilde{h}_* \circ i_*$. Since $\pi_1(\mathbb{R}^n, a_0)$ is the trivial group (as stated in Section 52 Example 1), $\tilde{h}_*: \pi_1(\mathbb{R}^n, a_0) \rightarrow \pi_1(Y, y_0)$ is the trivial homomorphism, hence so is $h_* = \tilde{h}_* \circ i_*$. $\square$

Problem 6 (Munkres 52.6). Let X be path connected. Let $h: X \rightarrow Y$ be continuous, with $h(x_0) = y_0$ and $h(x_1) = y_1$. Let α be a path in X from x_0 to x_1 , and let $\beta = h \circ \alpha$. Show that

$$\hat{\beta} \circ (h_{x_0})_* = (h_{x_1})_* \circ \hat{\alpha}.$$

Solution. Let $[f] \in \pi_1(X, x_0)$. Then,

$$(\hat{\beta} \circ (h_{x_0})_*)([f]) = [\bar{\beta}][h \circ f][\beta] = [h \circ \bar{\alpha}][h \circ f][h \circ \alpha] = (h_{x_1})_*([\bar{\alpha}][f][\alpha]) = ((h_{x_1})_* \circ \hat{\alpha})([f]). \quad \square$$